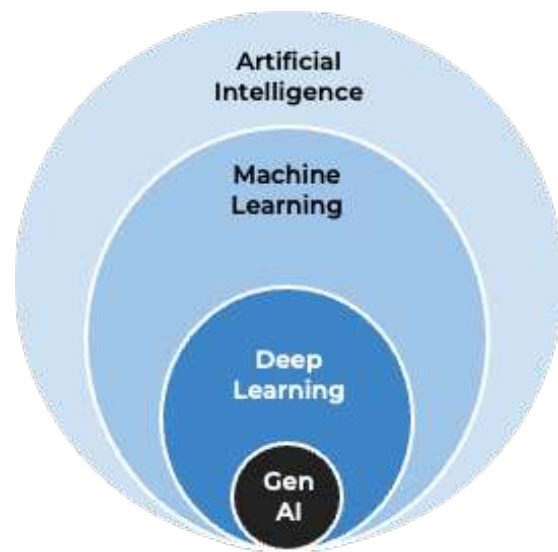


Generative AI

The Foundations

AI and Machine Learning

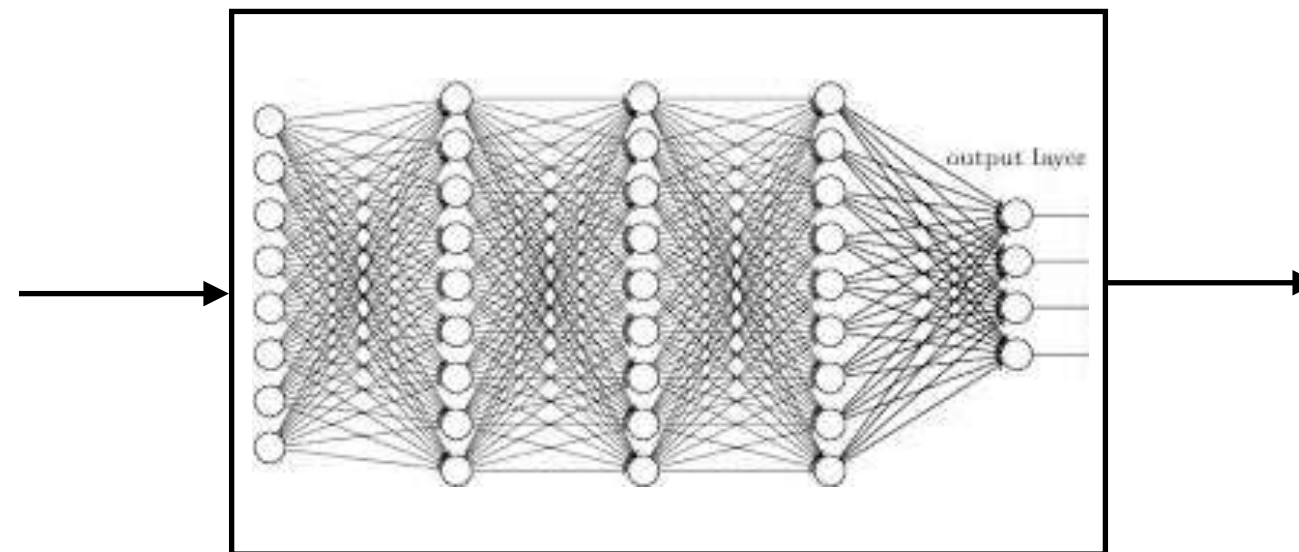


Discriminative AI vs Generative AI

- The Critic & The Artist

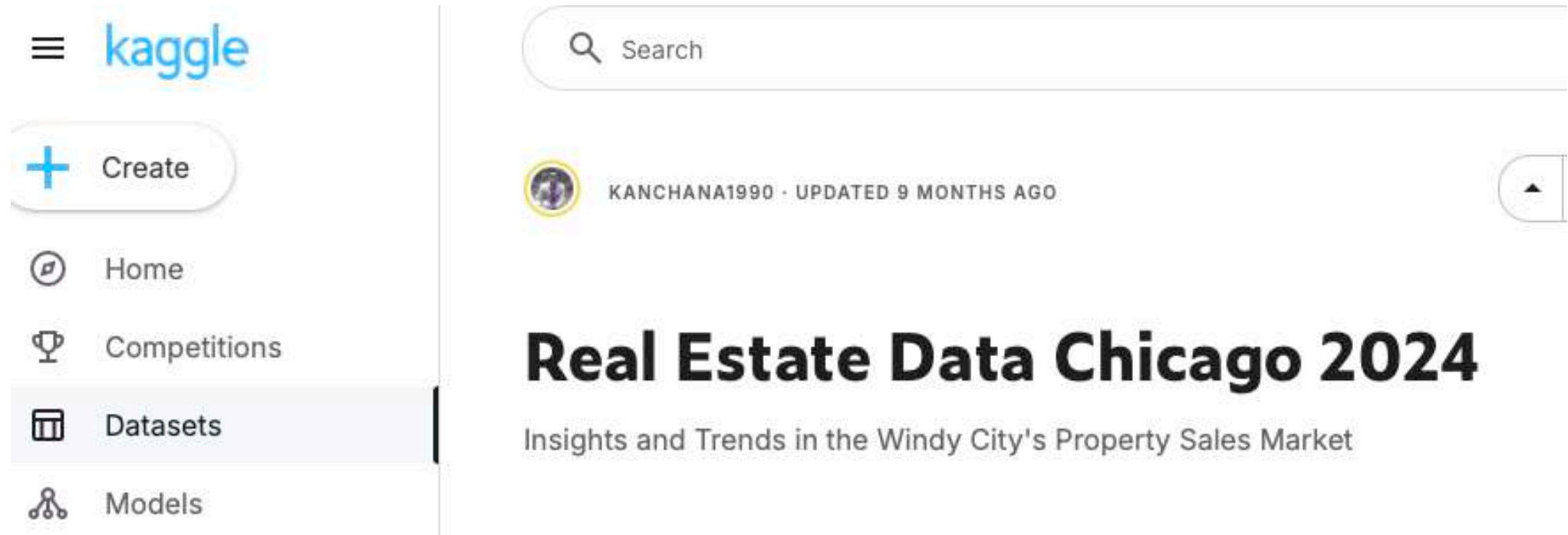
Large Language Models!

- Lets start with a very simple language model:



Large Language Models!

- Lets start with a very simple “small character model”
- Data



Come check out this amazing, move-in ready, 5 bed, 3.5 bath in the Beautiful Edgebrook of Chicago. Completely updated within the past few years with attention to detail. Quality is everywhere throughout this home including a finished basement with a suite-like style bedroom, walk-in closet and it's own bathroom. Wake up with a cup of coffee off of your balcony connected to the master bedroom. Granite in the kitchen, gorgeous cabinets and high end appliances make this kitchen a dream. Amazing hardwood floors throughout the home, a spacious living room, large windows, 2 wood burning fire places and plenty of hot...

Total 2 Million Characters

After 1 Epoch:

C _ _ _ _ _

cute cottage, clse to cllgs, crnr lctn, cvr prch, cgrg, 2 bd 1.5 bth...

chrnng condo, cty vws, crpt, clset spc, 3 bd 2 bth, ckitchen rnr unit...

cozy crnr lot, clse to cmm, 2 bd 1 bth, crpt flrs, lg bkyard, cntrl ac...

After 5 Epoch:

C _ _ _ _ _

cozy cnd home, clse to city, 3 bd 2 bth, covrd patio, centr ac, big bkyard...

cute cottage, clse to cty, 2 bd 1.5 bth, cvr prch, czl lng rm, updt flrs..

chrng cornr lot, cmmnty pool, 2 bd 2 bth, cpt flrs, updt kchn, convnt loc...

After 10 Epoch:

C _ _ _ _ _

charming cottage near city center, 2 bd 1.5 bth, covered porch, spacious living room, updated flooring...

cozy condo with updated kitchen, carpet floors, 2 bd 2 bth, community pool, convenient location near shops...

charming corner home, close to city, 3 bd 2 bth, cozy living area, covered patio, central hvac...

After 100 Epoch:

C _ _ _ _ _

charming corner lot home with 3 bedrooms and 2 bathrooms, featuring an open floor plan, updated kitchen with granite countertops, and a spacious backyard with a covered patio. conveniently located near shopping and schools.

cozy condo in a gated community, offering 2 bedrooms, 2 bathrooms, hardwood floors, and a modern kitchen with stainless steel appliances. community pool, fitness center, and easy access to public transportation.

cute craftsman-style home with a welcoming front porch, 4 bedrooms, 3 bathrooms, and a large fenced backyard. features include a fireplace in the living room, vaulted ceilings, and an attached two-car garage. close to parks and dining.

Ours vs GPT

	Ours	GPT
Type	Char Gen	Word Gen
Architecture	RNN	Transformer
Parameters	4 Million	2 Trillion
Training	1 Million Chars	10 Trillion Words
Cost	<\$1	>\$150 Million
Comp Time	10 mins x 1 Comp	150 days x10000 Comp